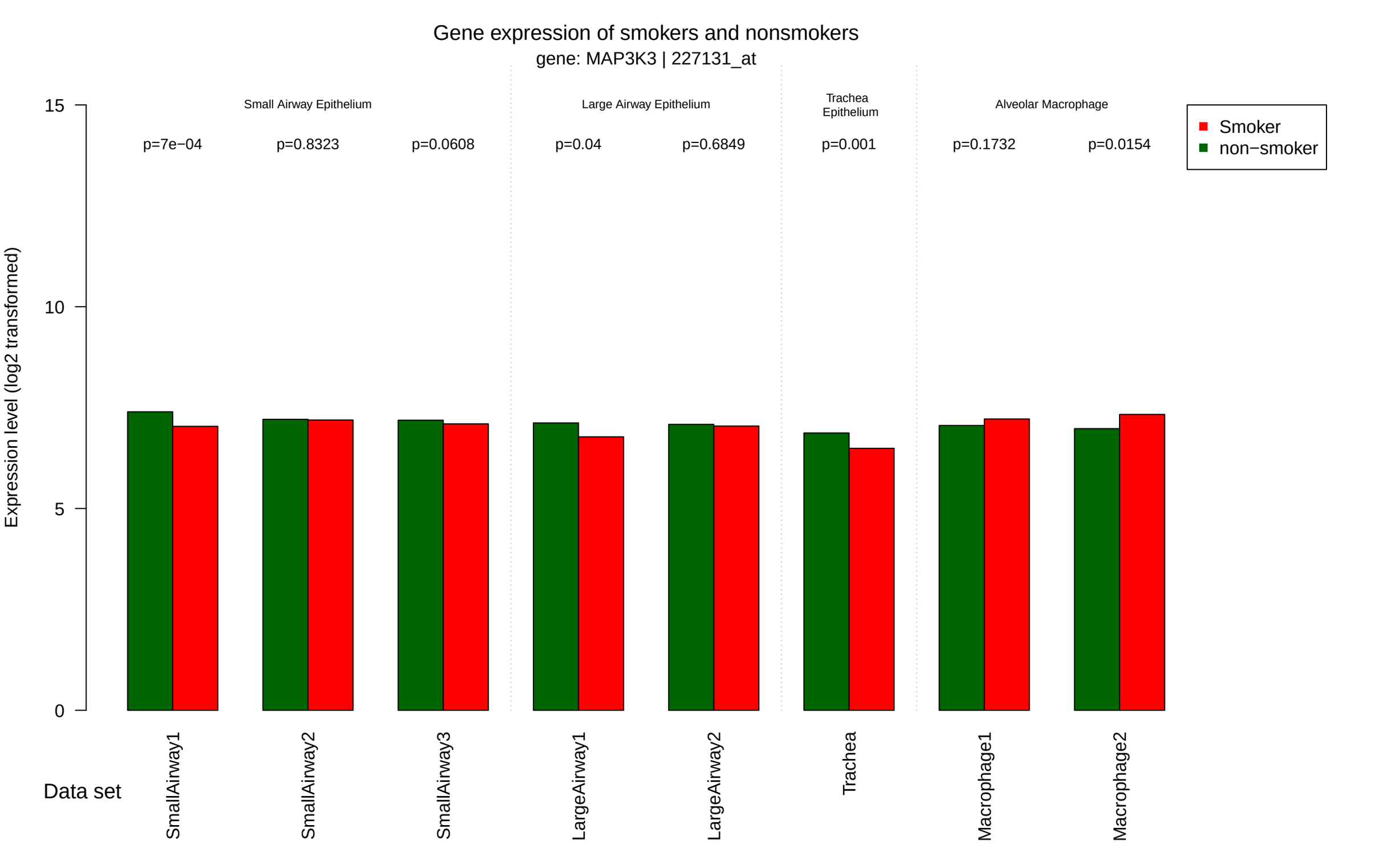
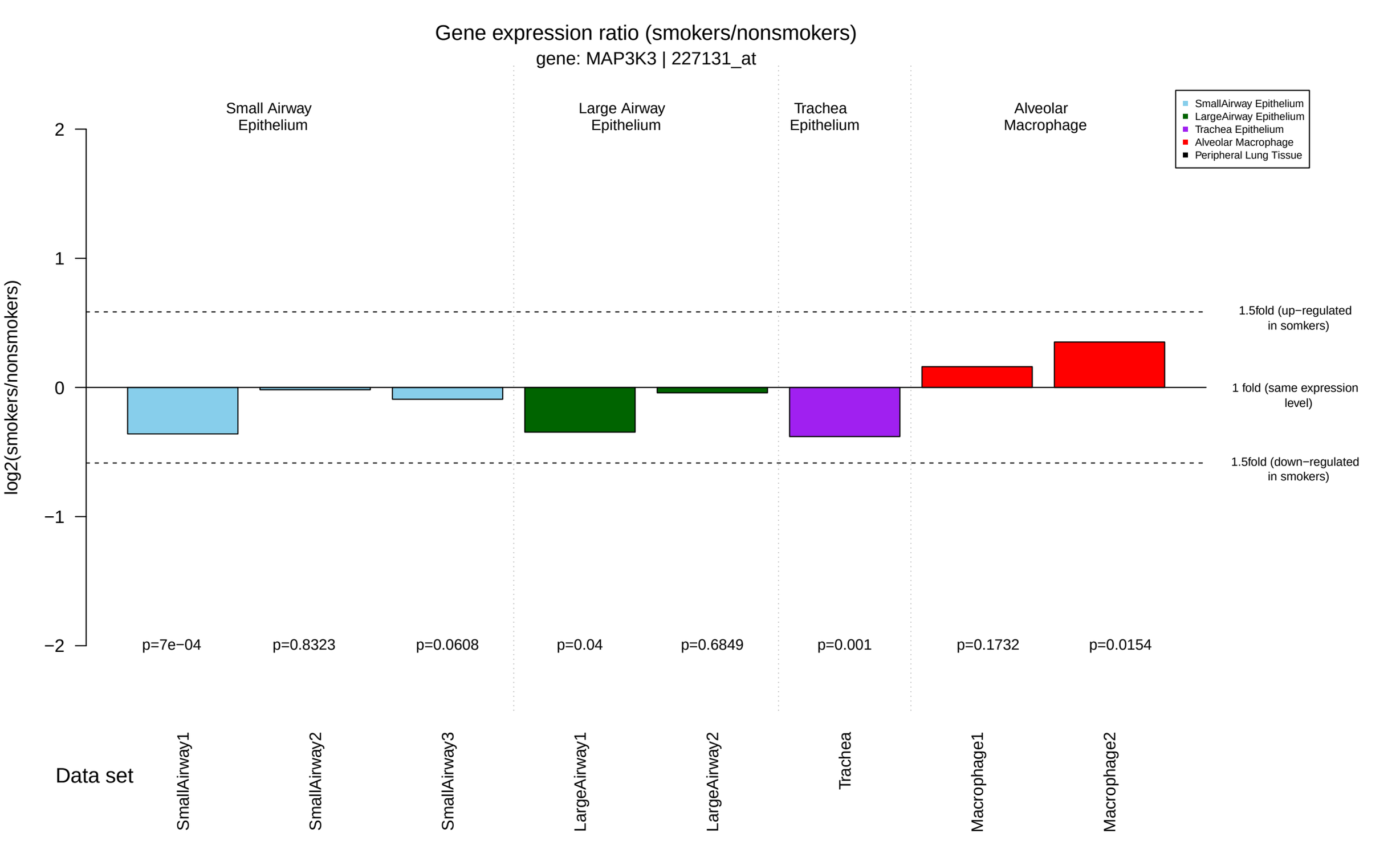
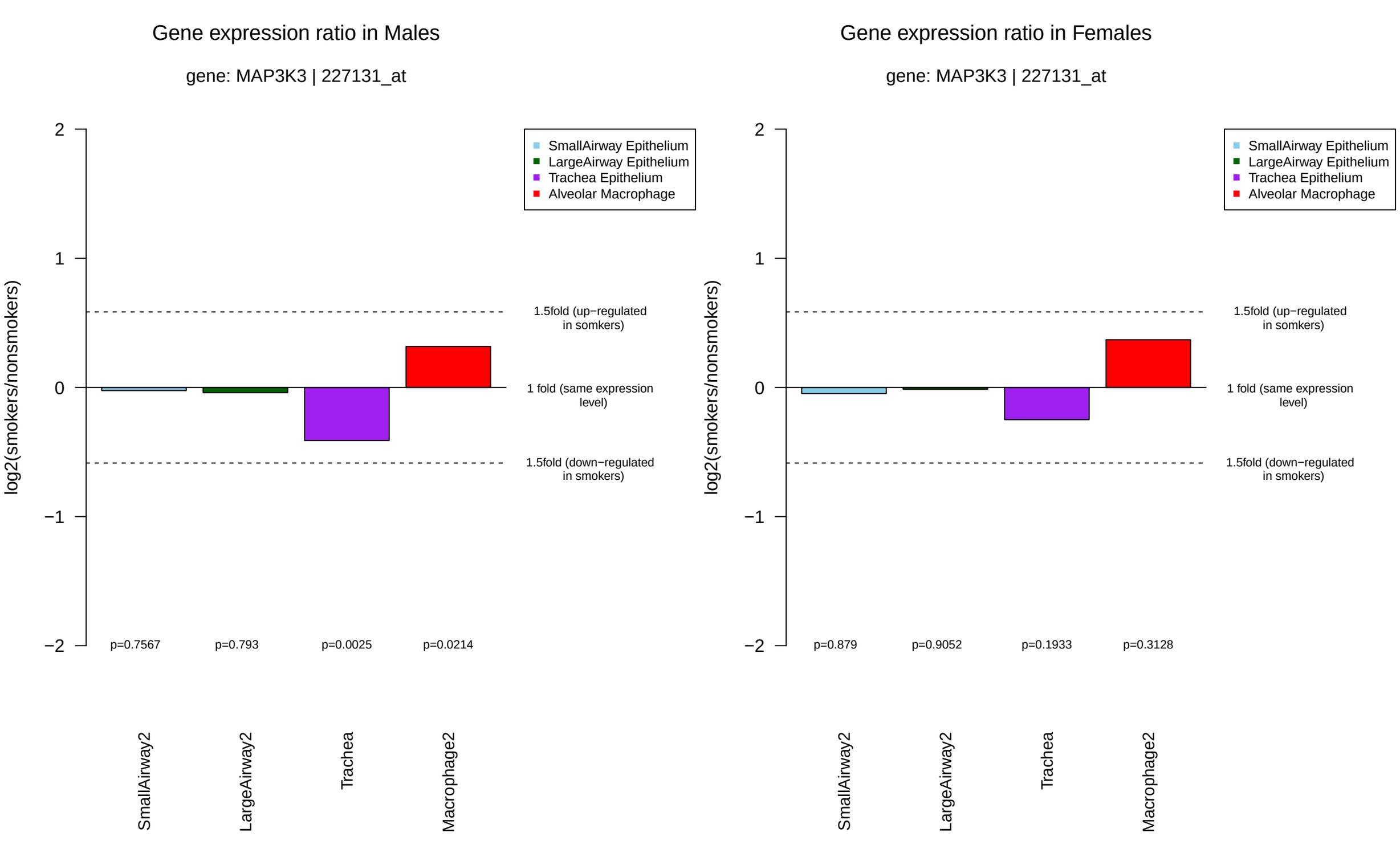
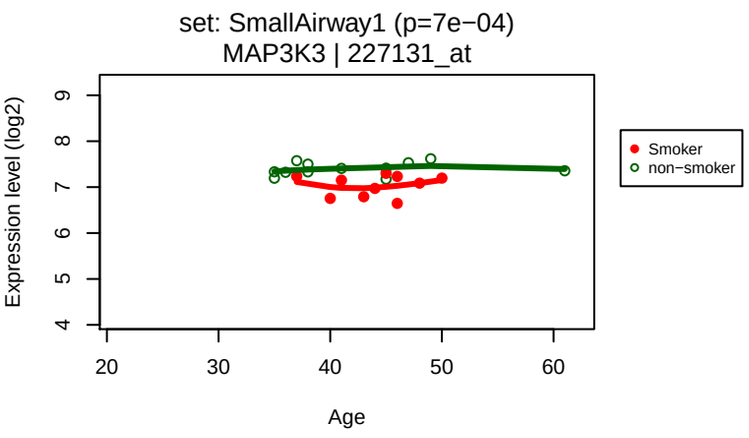
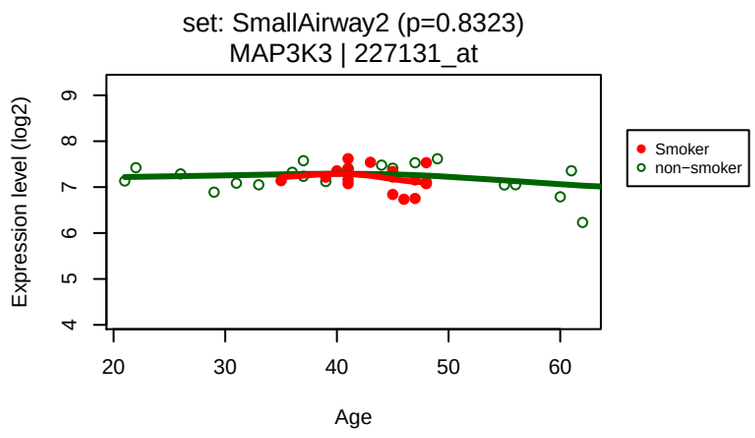
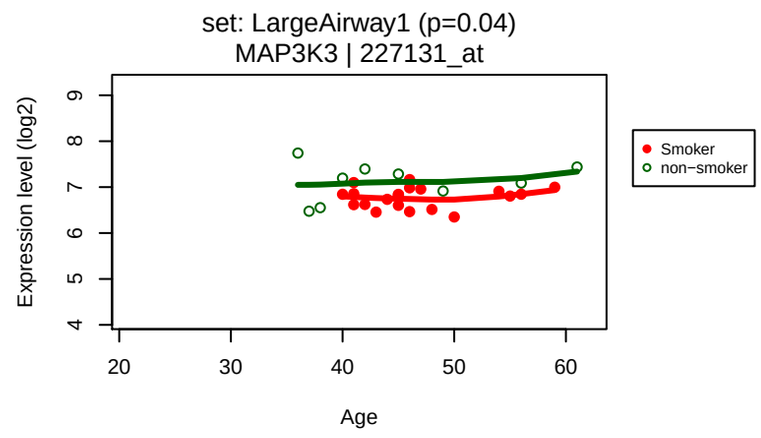

Gene expression of smokers and nonsmokers
gene: MAP3K3 | 227131_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=7e−04
p=0.8323
p=0.0608
p=0.04
p=0.6849
p=0.001
p=0.1732
p=0.0154
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

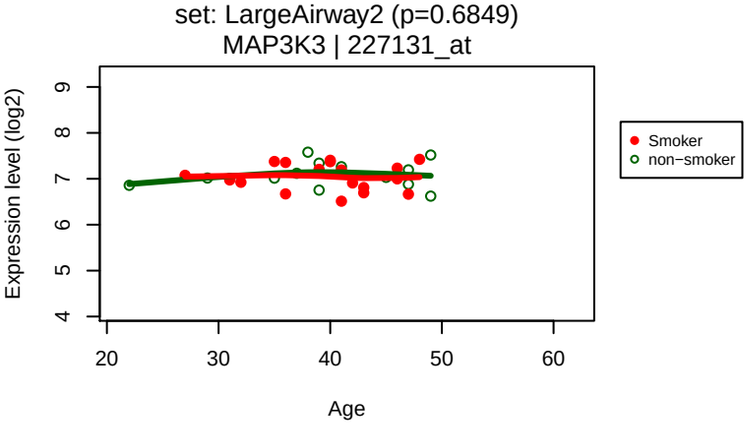

Gene expression ratio (smokers/nonsmokers)
gene: MAP3K3 | 227131_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=7e−04
p=0.8323
p=0.0608
p=0.04
p=0.6849
p=0.001
p=0.1732
p=0.0154
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

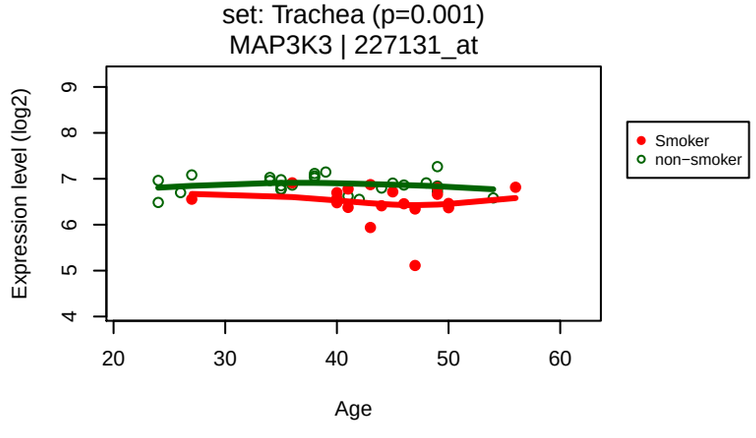

Gene expression ratio in Males
gene: MAP3K3 | 227131_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7567
p=0.793
p=0.0025
p=0.0214
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAP3K3 | 227131_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.879
p=0.9052
p=0.1933
p=0.3128
SmallAirway2
LargeAirway2
Trachea
Macrophage2

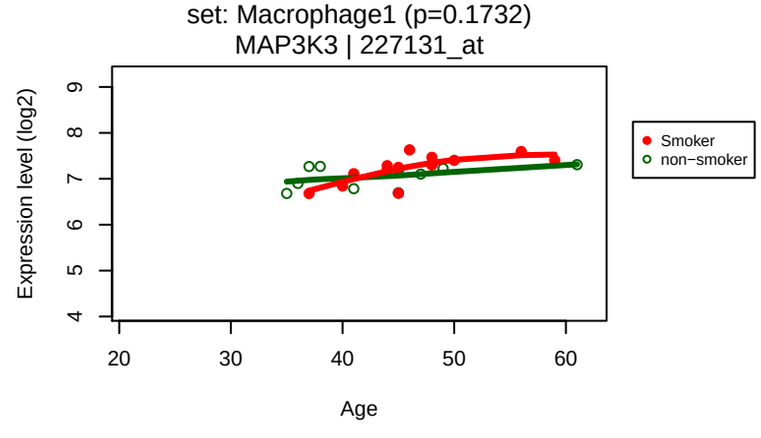

set: SmallAirway1 (p=7e−04)
MAP3K3 | 227131_at
Expression level (log2)
Age
Smoker
non−smoker

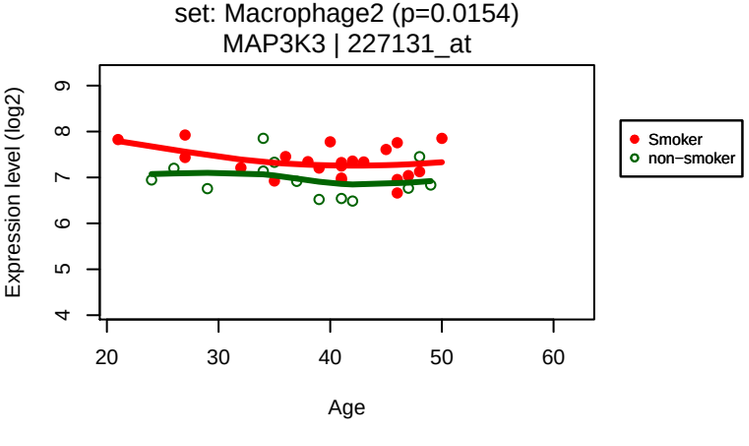

set: SmallAirway2 (p=0.8323)
MAP3K3 | 227131_at
Expression level (log2)
Age
Smoker
non−smoker

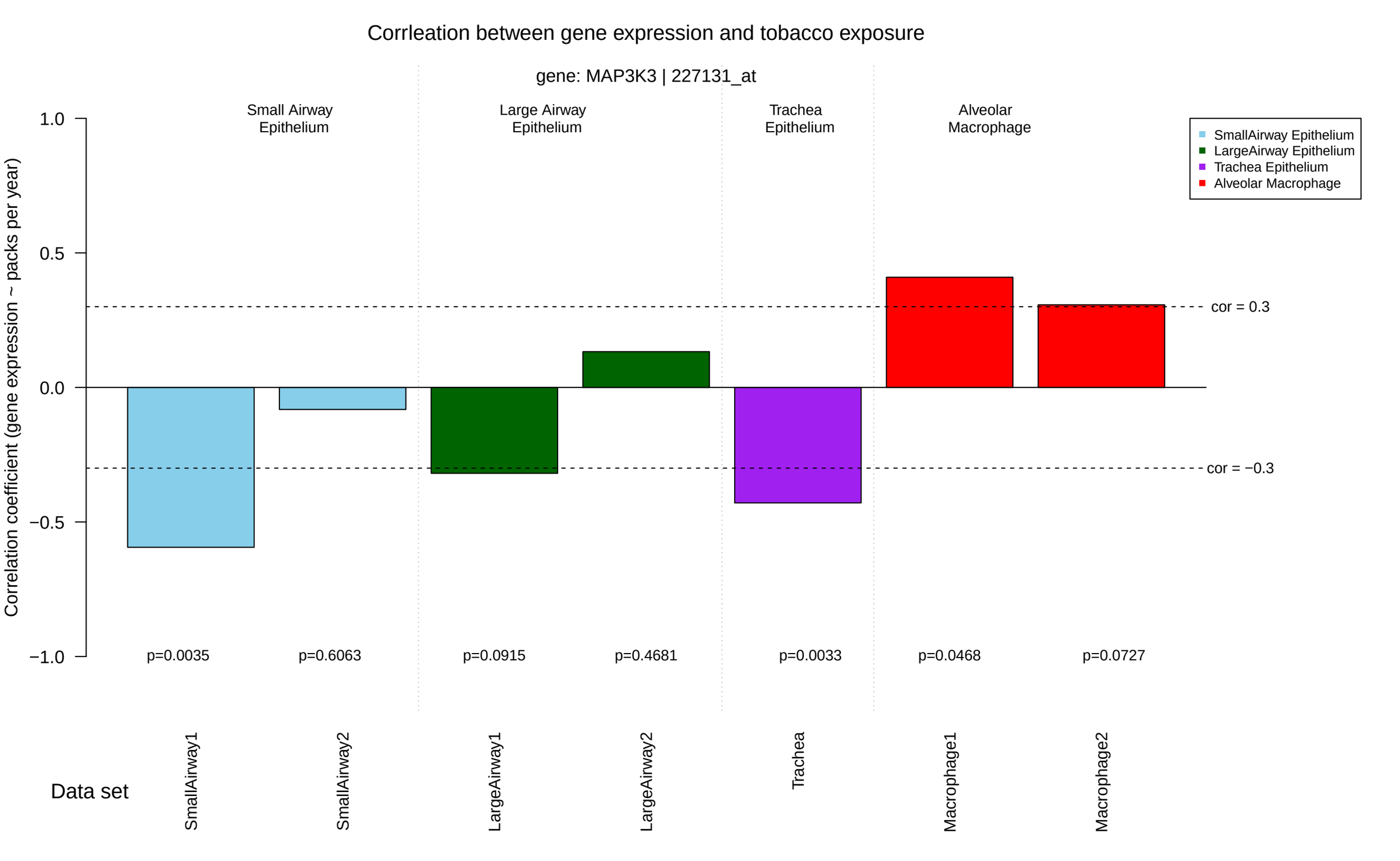

set: LargeAirway1 (p=0.04)
MAP3K3 | 227131_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6849)
MAP3K3 | 227131_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.001)
MAP3K3 | 227131_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1732)
MAP3K3 | 227131_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0154)
MAP3K3 | 227131_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAP3K3 | 227131_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0035
p=0.6063
p=0.0915
p=0.4681
p=0.0033
p=0.0468
p=0.0727
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2